HOLY ANGEL UNIVERSITY

IOT-BASED LABORATORY EQUIPMENT TRACKING FOR COMPUTING LABORATORIES

A Capstone Project
Presented to the Faculty of the
School of Computing
Holy Angel University

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Computer Science

Quebec Student
Romeo Student
Juliett Student
X-ray Student

November 2026

Final manuscript — revised after the final oral defense (see Summary of Revisions).
Dummy document generated for testing the HAU-SOC Thesis & Capstone prototype. All names, data and references are placeholders.

SCHOOL OF COMPUTING

APPROVAL SHEET

This capstone entitled **"IOT-BASED LABORATORY EQUIPMENT TRACKING FOR COMPUTING LABORATORIES"**, prepared and submitted in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science has been examined and is recommended for acceptance and oral examination.

Dean Delta
Adviser

ORAL EXAMINATION

Approved by the committee of oral examiners on
November 9, 2026

Prof. Foxtrot
Panel Chair

Prof. Bravo
Panel Member

[Name of Panel Member]
Panel Member

APPROVAL

Accepted and approved in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science.

Prof. Alpha
Program Chair/Coordinator,
Computer Science

Dean Delta
Dean, School of Computing

Acknowledgment

The researchers extend their sincere gratitude to everyone who supported this capstone project. This page is placeholder text: a real manuscript thanks the Adviser, the panel, the instructors, the host office and the families who made the work possible.

The group is especially grateful to Dean Delta for patient guidance through every revision.

Table of Contents

Page

Cover Page i
Approval Sheet ii
Acknowledgment iii
Table of Contents iv
Summary of Revisions vi
Title Page 1
Introduction 2
Review of Related Literature 2
- Indoor Positioning with Bluetooth Low Energy 2
- Asset Tracking in Educational Laboratories 2
- Signal Smoothing and Classification 3
- Evaluating IoT Systems 3
- Synthesis 3

Conceptual Framework 3
Objectives of the Study 3
Scope and Delimitations 4
Method 4
- Research Design 4
- Sources of Data 5
- Participants 5
- Instruments 6
- Data Collection 6
- Data Analysis 6
- Ethical Consideration 6

Results 6
Discussion 7
Conclusion 7
Recommendations 7
References 8
- Appendix A Cover Letter to Host/Locale 9

Appendix B Current Process Flowchart 10
Appendix C Functionality Checklist 11
Appendix D System Usability Scale Questionnaire 12
Appendix E ISO/IEC 25010 Software Quality Evaluation Form 13

Summary of Revisions

The following revisions were made in response to the panel's recommendations recorded in the Capstone Revision Form (FM-AAC-SOC-2004) after the final defense on November 9, 2026.

Panel recommendation	Action taken	Section
Add the per-item evaluation results to the Results, not only the means.	Results now include the full evaluation table per item.	Results
Explain the lowest-rated characteristic in the Discussion.	Discussion now explains the cause of the lowest rating and the fix made.	Discussion
Align the recommendations with the limitations found.	Each recommendation now answers a limitation found during testing.	Recommendations

Checked by: _____ Dean Delta, Adviser

IoT-Based Laboratory Equipment Tracking for Computing Laboratories

Quebec Student, Romeo Student, Juliett Student, X-ray Student

Holy Angel University
Computer Science

Abstract

This study aimed to design, build and evaluate an IoT system that locates tagged laboratory equipment by room and records its movement. Laboratory staff cannot locate equipment without searching every room, borrowing records are incomplete, and losses are discovered only at inventory time. The tracker located tagged equipment by room with 94.3% accuracy, completed the borrowing records, and was rated highly by laboratory staff and IT experts. Room-level accuracy was 94.3% (377 of 400 moves) with a 10-second smoothing window, and 86.8% without smoothing.

Keywords: capstone, web application, ISO/IEC 25010, usability

IOT-BASED LABORATORY EQUIPMENT TRACKING FOR COMPUTING LABORATORIES

Introduction

The computing laboratories lend projectors, development boards, networking kits and spare peripherals to classes every day. Borrowing is recorded on paper, items are moved between rooms without a record, and the semestral inventory regularly finds equipment missing or in the wrong room.

Bluetooth Low Energy (BLE) tags are inexpensive and last months on a coin cell. Gateways in each room can detect the tags and report their signal strength, from which a server can estimate the room an item is in. Combined with a borrowing log, such a system would show where every item is and flag items that leave the laboratories.

The problem this study addresses is the following: laboratory staff cannot locate equipment without searching every room, borrowing records are incomplete, and losses are discovered only at inventory time. This capstone project therefore proposes the equipment tracker.

Review of Related Literature

This section reviews studies and existing systems related to the equipment tracker and the methods used to evaluate it.

Indoor Positioning with Bluetooth Low Energy

RSSI-based room-level positioning with one receiver per room achieves over 90% accuracy when readings are smoothed over a short window (Placeholder & Author, 2023).

Asset Tracking in Educational Laboratories

Tagged asset systems in school laboratories reduced search time and made borrowing records complete (Sample, 2024).

Signal Smoothing and Classification

Moving averages and Kalman filters reduce RSSI noise; a nearest-gateway rule with hysteresis prevents items from flickering between rooms (Example & Writer, 2022).

Evaluating IoT Systems

ISO/IEC 25010 is commonly applied to IoT prototypes alongside accuracy measurements of the sensing layer (Example, 2023).

Synthesis

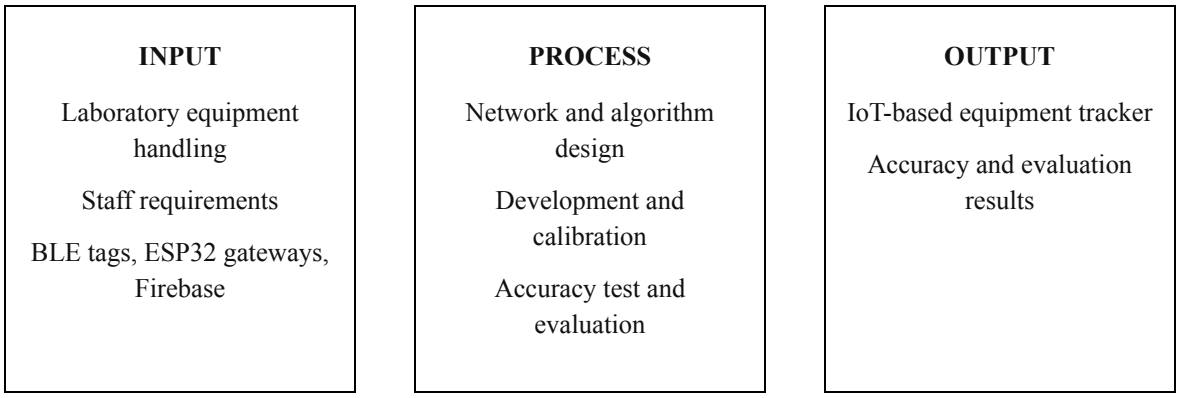
The reviewed studies show that the problem is common and that a focused, web-based solution can address it at low cost. None of the systems reviewed fits the setting of this study exactly, which is the gap the equipment tracker fills.

Conceptual Framework

The study follows the Input–Process–Output model shown in Figure 1.

Figure 1

Conceptual Framework of the Study



Objectives of the Study

The general objective of the study is to design, build and evaluate an IoT system that locates tagged laboratory equipment by room and records its movement. Specifically, the study aims to:

1. determine the equipment-handling problems of the computing laboratories through observation and interviews;
2. design a BLE tag and gateway network and a room-level location algorithm based on received signal strength;
3. develop a web dashboard that shows the location and movement history of each item and alerts staff when an item leaves the laboratories;
4. measure the room-level location accuracy of the algorithm in the four computing laboratories; and
5. evaluate the system using ISO/IEC 25010 with laboratory staff and IT experts.

Scope and Delimitations

The study covers the four computing laboratories and the equipment room on one floor, using BLE tags on 60 items and one gateway per room. Location is resolved to the room, not to a position within it.

The study does not track people, does not cover equipment outside the floor, and does not replace the university's property inventory system; it provides location and movement records that staff can reconcile with it.

Method

This section describes how the equipment tracker will be designed, built, measured and evaluated.

Research Design

The study uses a developmental research design with a quantitative evaluation. The location algorithm is developed iteratively and its accuracy is measured experimentally.

Table 1

Functional Requirements

ID	User	Requirement

FR-01	Staff	Register an item and attach a BLE tag
FR-02	System	Estimate the room of each tagged item every five seconds
FR-03	Staff	View the location and movement history of an item
FR-04	System	Alert staff when an item leaves the laboratories
FR-05	Staff	Record borrowing and return of an item
FR-06	Supervisor	Generate inventory and movement reports

Table 2

Non-Functional Requirements

Characteristic	Requirement
Accuracy	Room-level location is correct for at least 90% of moves
Performance	A move appears on the dashboard within ten seconds
Reliability	A gateway that goes offline is reported within one minute
Security	Only laboratory staff accounts can view locations

Sources of Data

RSSI readings from 60 tagged items collected by four gateways every second for two weeks, and ground-truth room labels recorded by the researchers during 400 controlled moves.

Participants

Five laboratory staff and five IT experts evaluate the system; no personal data about students is collected.

Table 3

Respondents of the Study

Respondent group	Number	Instrument
Laboratory staff	5	ISO/IEC 25010
IT experts	5	ISO/IEC 25010

Instruments

The ISO/IEC 25010 evaluation form (Appendix H) and an accuracy test log recording the true and predicted room of each move.

Data Collection

After approval from the Laboratory Supervisor, the researchers install the gateways, tag the items, and collect readings for two weeks while recording controlled moves.

Data Analysis

Accuracy is the share of moves whose predicted room matches the true room after the smoothing window. ISO/IEC 25010 ratings are summarized by weighted mean (Table 6).

Ethical Consideration

Participation is voluntary and respondents may withdraw at any time. Responses are anonymous, kept for the duration of the study and destroyed after the final defense, in accordance with the Data Privacy Act of 2012.

Results

Table 4

Functionality Checklist Results

Function	Result
Register item and tag	Working
Room estimate every five seconds	Working
Location history	Working
Out-of-laboratory alert	Working
Borrowing and return	Working
Inventory report	Working
Gateway offline alert	Working

Table 5

ISO/IEC 25010 Evaluation

Characteristic	Weighted mean	Interpretation
Functional Suitability	4.46	Strongly Agree
Performance Efficiency	4.20	Agree
Usability	4.36	Strongly Agree
Reliability	4.12	Agree

Location accuracy. Room-level accuracy was 94.3% (377 of 400 moves) with a 10-second smoothing window, and 86.8% without smoothing.

Discussion

Smoothing over ten seconds raised accuracy above the 90% target at the cost of a short delay before a move appears, which staff found acceptable. Errors concentrated at the doorway between Laboratories 2 and 3, where both gateways receive similar signal strength; hysteresis reduced but did not remove these errors.

Conclusion

The tracker located tagged equipment by room with 94.3% accuracy, completed the borrowing records, and was rated highly by laboratory staff and IT experts.

Recommendations

1. Add a second gateway at shared doorways.
2. Integrate with the university property inventory to reconcile records automatically.
3. Test tag battery life over a full semester.

References

Author, A. B., Writer, C. D., & Name, E. F. (2023). A placeholder study of progressive web applications for campus services. Journal of Sample Computing, 12(3), 45–58.

Example, G. (2023). Evaluating capstone systems against a software quality model: A placeholder review. Philippine Journal of Placeholder Studies, 8(1), 1–14.

Example, H., & Writer, I. (2022). Models for small service systems (placeholder ed.). Sample Press.

Placeholder, J. (2020). Measuring usability with a ten-item scale: A placeholder overview. Sample Usability Review, 5(2), 20–31.

Placeholder, K. (2021). The psychology of waiting revisited. Placeholder Service Quarterly, 30(4), 101–115.

Placeholder, L., & Author, M. (2023). Room-level positioning and scheduling in placeholder settings. Journal of Sample Operations, 3(1), 7–19.

Sample, N. (2022). Reminder timing and attendance: A placeholder study. In Proceedings of the Placeholder Conference on Information Systems (pp. 88–95). Sample Publisher.

Sample, O. (2024). Asset tracking in teaching laboratories. Sample Journal of Educational Technology, 2(2), 60–72.

Writer, Q. (2022). Choosing notification channels for low-budget systems. Placeholder Informatics, 9(1), 33–41.

Appendix A

Cover Letter to Host/Locale

February 16, 2026

The Laboratory Supervisor
Computing Laboratories Office
Holy Angel University

Dear Sir:

We are students of the School of Computing working on our capstone project entitled “IoT-Based Laboratory Equipment Tracking for Computing Laboratories.” We respectfully ask permission to observe the current process at your office, interview your staff, and conduct a test session of the system. This letter is a placeholder for the dummy manuscript.

Respectfully yours,

Quebec Student
Romeo Student
Juliett Student
X-ray Student

Noted by:
Dean Delta, Adviser

Appendix B

Current Process Flowchart

Request is made by message or on paper

Staff checks availability manually

Request is confirmed or declined

Activity takes place

Record is kept on paper, if at all

Appendix C

Functionality Checklist

Function	Working	Not working	Remarks
Register item and tag			
Room estimate every five seconds			
Location history			
Out-of-laboratory alert			
Borrowing and return			
Inventory report			
Gateway offline alert			

Appendix D

System Usability Scale Questionnaire

1 – Strongly Disagree, 2 – Disagree, 3 – Neutral, 4 – Agree, 5 – Strongly Agree

Statement	1	2	3	4	5
1. I think that I would like to use this system frequently.					
2. I found the system unnecessarily complex.					
3. I thought the system was easy to use.					
4. I think that I would need the support of a technical person to be able to use this system.					
5. I found the various functions in this system were well integrated.					
6. I thought there was too much inconsistency in this system.					
7. I would imagine that most people would learn to use this system very quickly.					
8. I found the system very cumbersome to use.					
9. I felt very confident using the system.					
10. I needed to learn a lot of things before I could get going with this system.					

Appendix E

ISO/IEC 25010 Software Quality Evaluation Form

Criteria	5	4	3	2	1
Functional Suitability					
The system meets the functional suitability requirements.					
Performance Efficiency					
The system meets the performance efficiency requirements.					
Usability					
The system meets the usability requirements.					
Reliability					
The system meets the reliability requirements.					